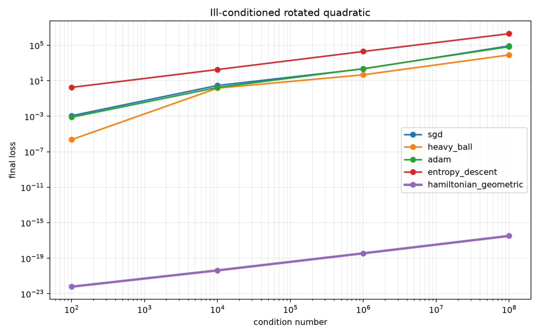
Hamiltonian–Geometric Optimization

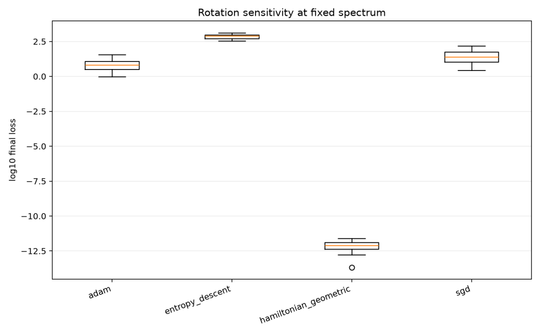
Optimization as motion on a curved landscape

Loss energy | **Metric geometry** | **Momentum state** | **Damping training** | **Evidence plots**

Not a magic optimizer: a geometry-first framework for structured learning problems.

Why Build This?





Core bet: use geometry instead of pretending the landscape is flat.

Core Idea

θ
position

$L(\theta)$
potential

$g(\theta)$
metric

p
momentum

γ
damping

$$H(\theta, p) = \frac{1}{2} p^\top g^{-1}(\theta) p + L(\theta)$$

Hamiltonian motion + dissipation = optimizer trajectory.

Derived vs Proposed

STANDARD

Hamilton's equations
Legendre transform
Rayleigh damping
Natural gradient

OUR SYNTHESIS

loss metric
memory force
spectral term
energy safeguard

HONEST CLAIM

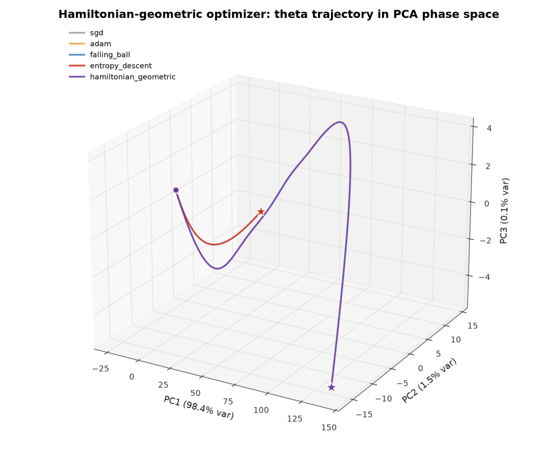
If no exact reference exists, call it proposed here.

Optimizer Family Tree

GD flat	**Momentum** inertia	**Adam** diagonal	**Natural** full metric	**HG** metric + p

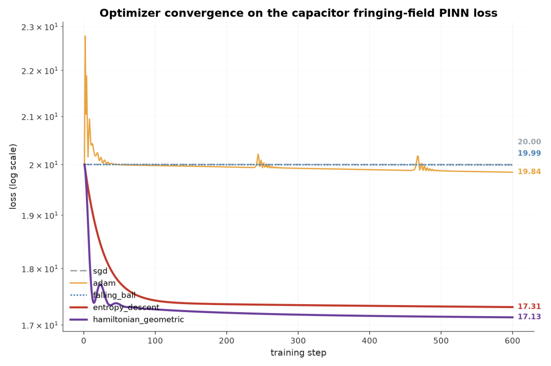
Correspondence, not historical authorship.

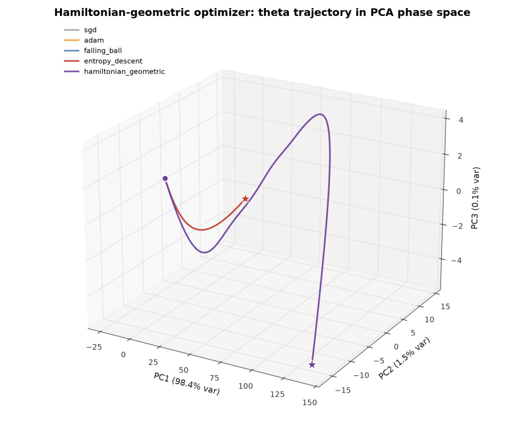
Motion Matters



Optimizer behavior is a trajectory in phase space, not only a final loss.

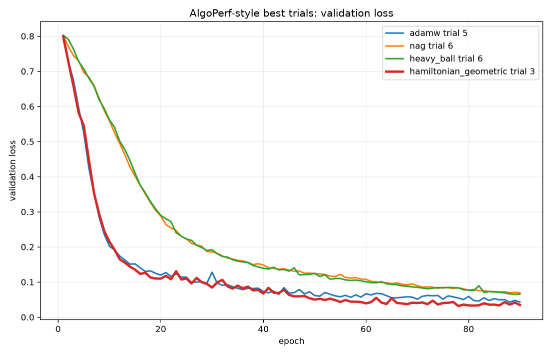
Physics-Informed Learning

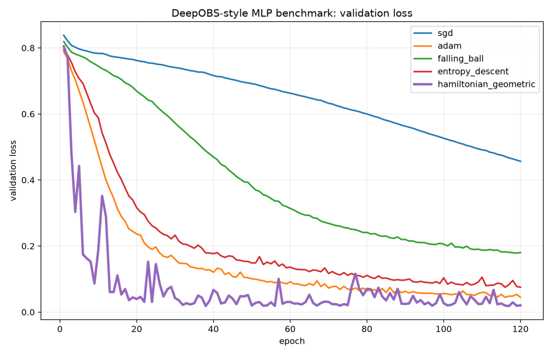




Strongest current story: structured physics loss + metric-aware motion.

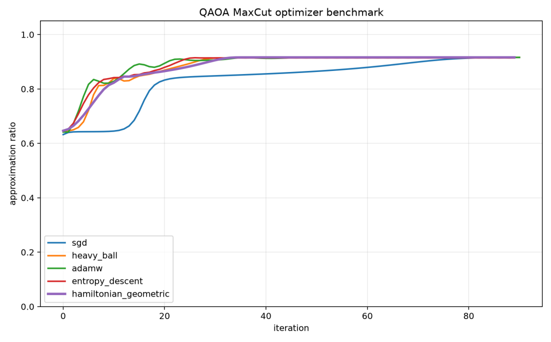
Training Dynamics

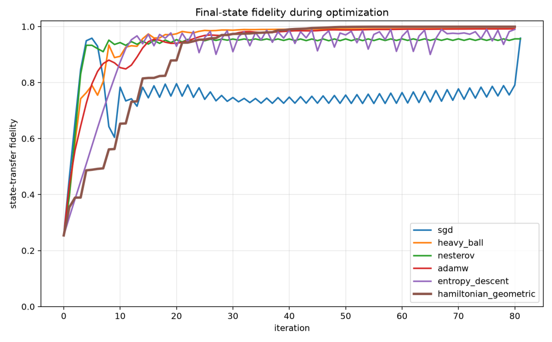




Convergence behavior is easier to explain visually than through tables.

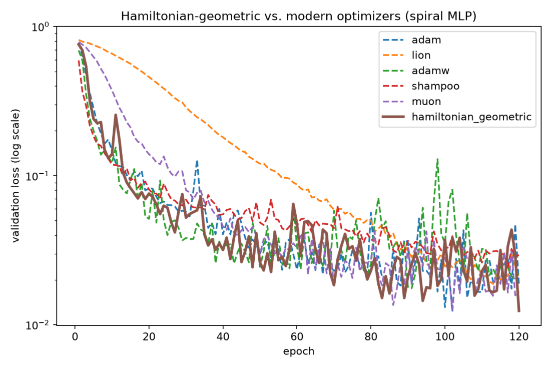
Quantum Optimization Evidence

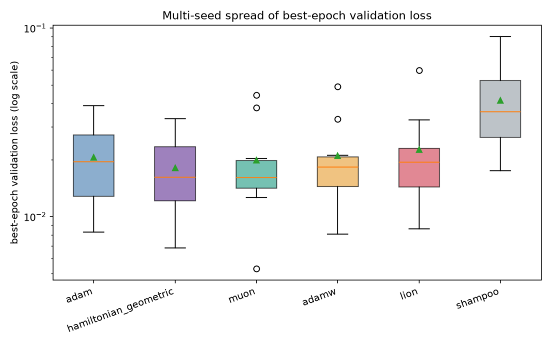




This is the bridge to spin chains and tensor networks.

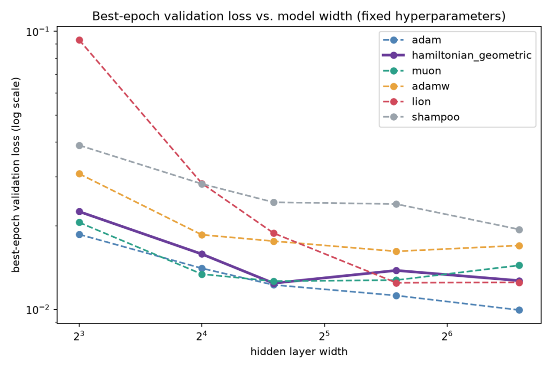
Modern Optimizers

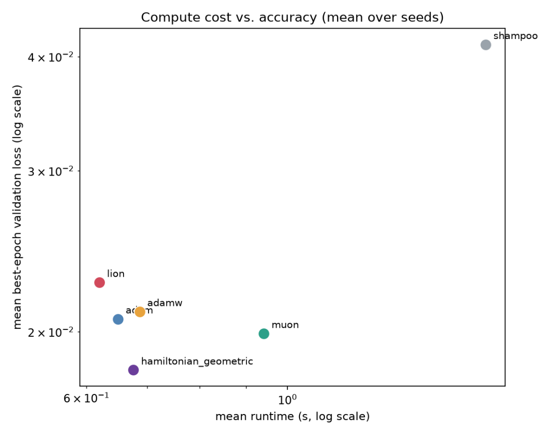




Honest result: Adam, HG, and Muon are close; seed variance matters.

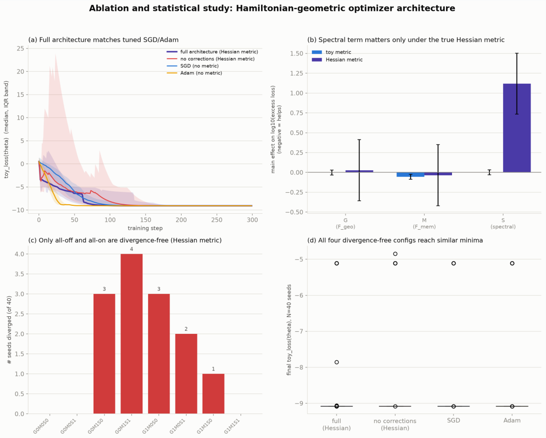
Scaling And Cost

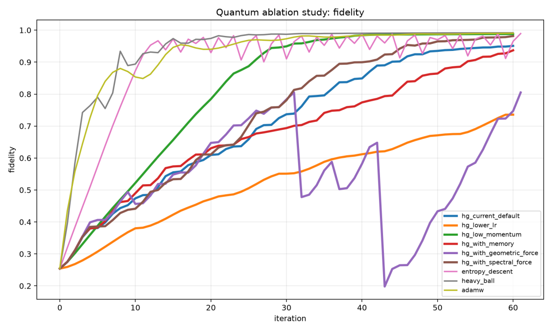




Do not oversell: geometry helps when structure justifies the cost.

What We Have Done





Current status: working prototype and evidence, not final theory.

Next: Tensor Networks

MPS state

local envs

block metric

HG step

gauge check

Gauge freedom   Bond dimension   Entanglement spectrum   C

First integration target: MPS/MPO variational optimization.

Next: Spin Chains

Ising | XXZ | J_1–J_2 | VQE | TDVP

Measure: energy, fidelity, symmetry stability, entanglement growt

First experiment: transverse-field Ising chain with MPS baseline.

Higher-Order Targets

PDE learning

Neural ODEs

Meta-learning

Quantum circuits

Implicit layers

Inverse problems

Rule: if curvature, constraints, or symmetries matter, try the geometric optimizer.

Roadmap

clean API

spin-chain benchmark

tensor backend

runtime normalization

publish carefully

North star: structured scientific and quantum optimization.

Attribution Rule

Derived from standard mechanics

equations

Proposed here as optimizer package

design + evidence

If no exact prior reference exists, say it plainly.